



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO S V V TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

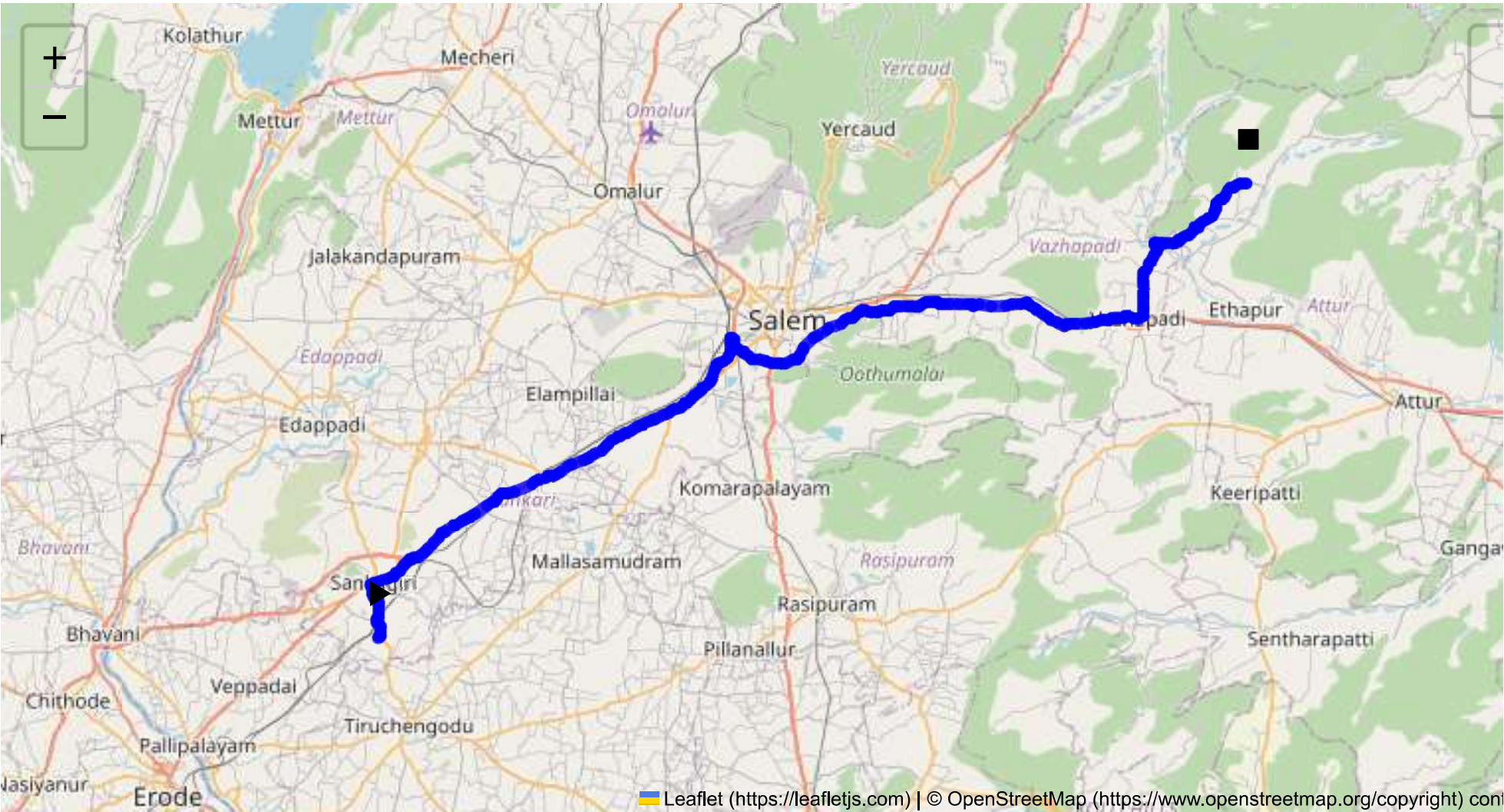
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 88.30 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.3 hours
Start: (11.4381, 77.8734)
End: (11.74835, 78.47672)



Welcome to the Journey Risk Management Study

To provide a comprehensive safety analysis of the route from CVQF+23W, Sangagiri to PFXG+8M Tumbal R.F. in Tamil Nadu, India, the following points have been assessed based on available information:

1. Overview of the Route Map:

- The route covers approximately 88.30 kilometers, primarily traversing through the National and State Highways between Sangagiri and Tumbal R.F. It includes a mix of urban and rural roads,

with varying degrees of infrastructure development.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu experiences a tropical climate with significant variations. The region is prone to monsoon rains from October to December, which can lead to waterlogging and slippery roads. Heatwaves are common from March to June and may affect vehicle performance.

3. Analysis of Traffic Patterns:

- Peak traffic hours generally occur between 8:00-10:00 AM and 5:00-8:00 PM. Congestion is typically observed near major towns and highway intersections, such as Erode and Salem Bypass.

4. Assessment of Road Quality and Infrastructure:

- Road conditions may vary, with National Highways typically in better shape compared to smaller rural roads. Potholes and uneven surfaces may be present, especially in less maintained rural sections.

5. Suggestions for Alternative Routes for Emergencies:

- In case of an emergency or blockage, an alternative route could be via NH44 connecting to other regional roads toward the destination. However, the availability of such routes should be verified in real-time due to potential local disruptions.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Regulations require proper documentation and adherence to designated transport corridors for hazardous materials. Vehicles must comply with weight limits and display appropriate signage.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- Past incidents along similar routes often involve vehicular collisions due to congestion or weather conditions. Regular monitoring of traffic advisories is recommended.

8. Environmental Considerations and Sensitive Areas:

- The route may pass near environmentally sensitive areas such as agricultural lands and water bodies. Adhering to spill management protocols is crucial to prevent environmental impact.

9. Analysis of Communication Coverage:

- Most of the route should have adequate mobile network coverage, but remote or rural segments may experience temporary dead zones. Communication devices should be fully charged or backed by satellite support where possible.

10. Estimated Emergency Response Times for Different Route Segments:

- Response times can vary, with quicker responses likely near urban centers like Erode. Remote areas may take longer, potentially impacting timely medical or mechanical assistance.

11. Overall Summary of Risk Assessment:

- While the route is generally manageable with proper precautions, key risks include adverse weather conditions, traffic congestion, and road quality issues. Preparation and continuous monitoring of local conditions and advisories are vital. Ensuring all safety protocols, such as vehicle maintenance and driver readiness, can help mitigate potential risks effectively.

This assessment highlights key safety considerations and emphasizes the importance of understanding dynamic conditions and local regulations while transporting hazardous materials along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45000, 77.87213	30 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.45309, 77.87325	30 KM/Hr
7	Turn	Medium	11.46400, 77.87323	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr
12	Turn	Medium	11.64172, 78.11817	30 KM/Hr
13	Turn	Medium	11.64216, 78.11875	30 KM/Hr
14	Turn	Medium	11.64195, 78.12129	30 KM/Hr
15	Turn	Medium	11.64162, 78.12160	30 KM/Hr
16	Turn	Medium	11.64120, 78.12168	30 KM/Hr
17	Turn	Medium	11.63948, 78.12081	30 KM/Hr
18	Turn	Medium	11.65459, 78.37964	30 KM/Hr
19	Turn	Medium	11.65472, 78.37991	30 KM/Hr
20	Turn	Medium	11.65484, 78.38000	30 KM/Hr
21	Turn	Medium	11.65602, 78.38711	30 KM/Hr
22	Turn	High	11.65716, 78.40570	15 KM/Hr
23	Turn	Medium	11.65743, 78.40584	30 KM/Hr
24	Turn	High	11.70447, 78.41562	15 KM/Hr
25	Turn	Medium	11.70480, 78.41538	30 KM/Hr
26	Turn	High	11.70565, 78.41336	15 KM/Hr
27	Turn	High	11.70684, 78.41358	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
28	Turn	Medium	11.71553, 78.44012	30 KM/Hr
29	Turn	Medium	11.72306, 78.44868	30 KM/Hr
30	Turn	Medium	11.72620, 78.45297	30 KM/Hr
31	Turn	Medium	11.72682, 78.45305	30 KM/Hr
32	Turn	Medium	11.74455, 78.46603	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
22	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
23	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
24	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
25	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
26	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
27	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
28	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
29	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
30	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
31	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
35	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
37	hospital	Niharika Hospital	11.6565971, 78.3937356	30 km/h	Medium
38	hospital	Malar Hospital	11.6567854, 78.3928345	30 km/h	Medium
39	hospital	Sri Bala Hospital	11.6566667, 78.4010889	30 km/h	Medium
40	clinic	Sibi Physiotherapy Clinic	11.6561305, 78.3994263	30 km/h	Medium
41	hospital	Saraswathi Hospital, Valapady	11.6563505, 78.4009062	30 km/h	Medium
42	hospital	Devaki Hospital	11.6564189, 78.402376	30 km/h	Medium
43	hospital	Sri Udaya Hospital	11.6560244, 78.4040862	30 km/h	Medium
44	hospital	Government Hospital, Valapady	11.65706, 78.403773	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
45	hospital	Deepam Hospital	11.6576703, 78.4065155	30 km/h	Medium
46	hospital	Government hospital,Belur	11.70532, 78.419538	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
21	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
32	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
33	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
34	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
36	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

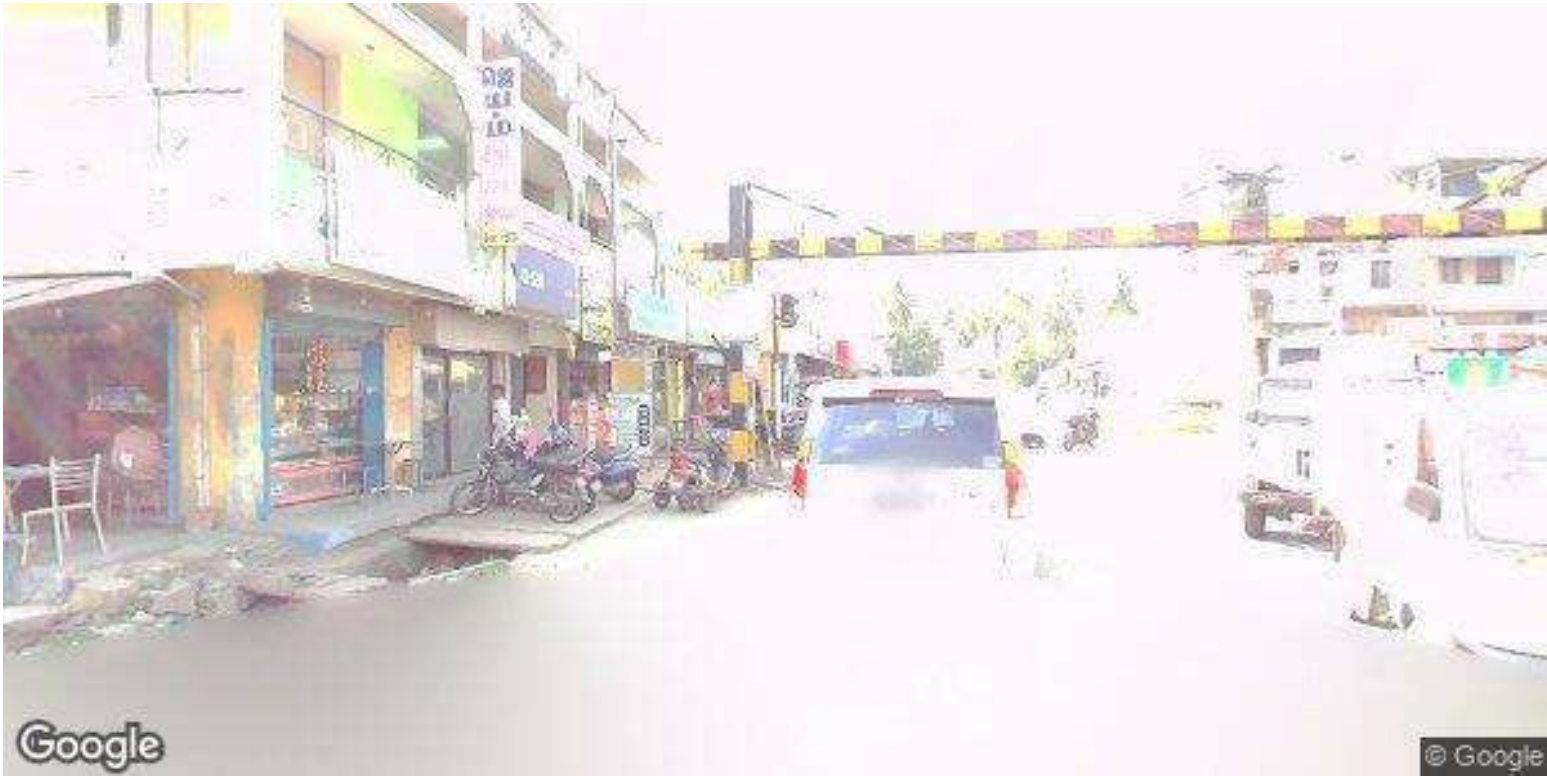
Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64172, 78.11817



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64216, 78.11875



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64195, 78.12129



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64162, 78.12160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64120, 78.12168



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63948, 78.12081



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65459, 78.37964



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65472, 78.37991



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65484, 78.38000



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65602, 78.38711



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65716, 78.40570



Coordinates: 11.65743, 78.40584



Coordinates: 11.70565, 78.41336



Coordinates: 11.70684, 78.41358



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71553, 78.44012



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72306, 78.44868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72620, 78.45297



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72682, 78.45305



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74455, 78.46603

